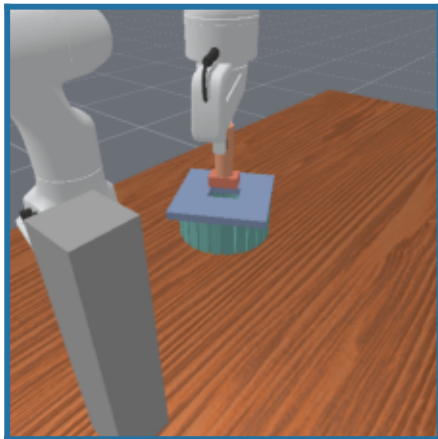


Keyed-to-circular peg insertion — real simulation scenes

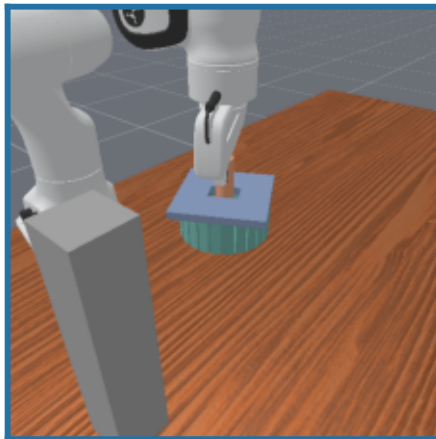
ManiSkill / PhysX · Panda pd_joint_pos · intervention $c = [\Delta x, \Delta y, \Delta \psi]$
frozen 39-condition manifest (30 mixed + 8 isolated + 1 zero) · held-out socket yaw +30° shown

Phase switch
(keyed task)

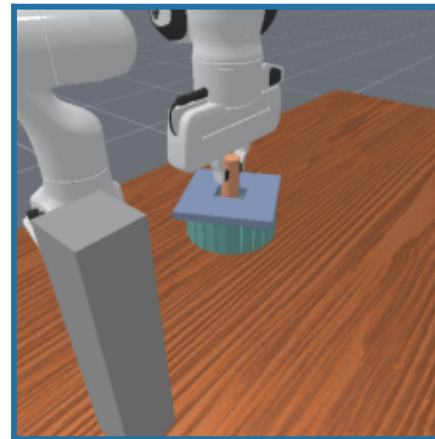
align_keyed



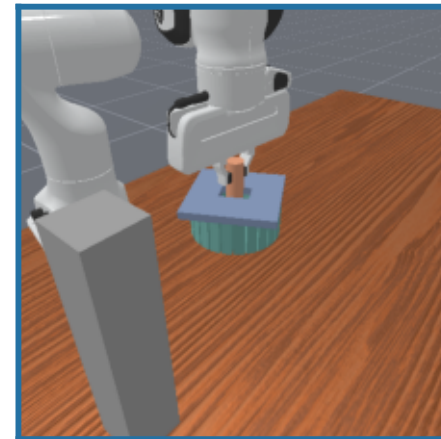
enter_key



unlock_yaw

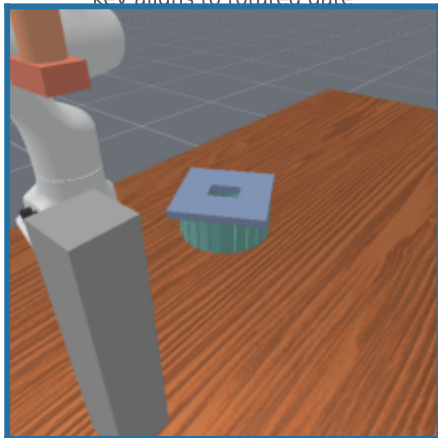


circular_insert



Keyed Q_1 (vertical)

key aligns to rotated date



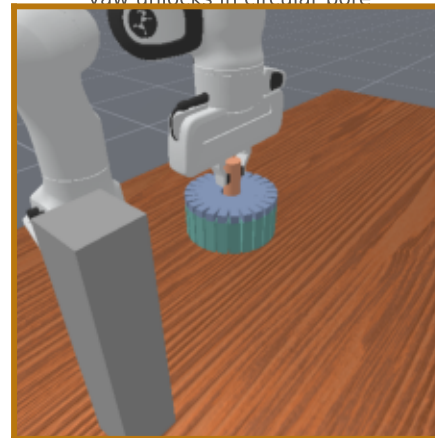
Keyed Q_2 (horizontal)

key passes through keyed date



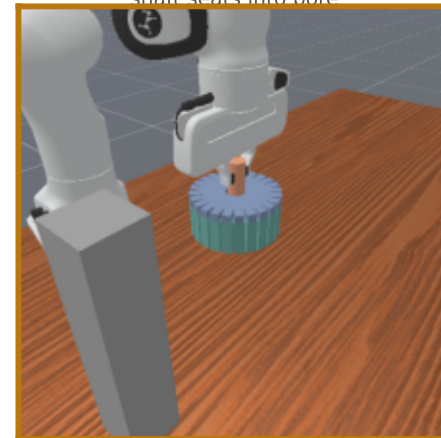
Circular honest

yaw unlocks in circular bore



Circular placebo

shaft seats into bore



single-seed 39 · five-seed 195
few-shot 1,815 fits · TP-GMM 80 fits

rotated axis 232 usable

circular symmetry 117 usable

circular symmetry 117 usable

6 experiments · 700 usable physics rollouts · 1895 offline model fits

all rollouts executed in simulation (ManiSkill/PhysX); failed attempts retained with condition id, attempt id, seed, stop reason.